

WASP-50b

R-1097

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_151228 · created 2026-07-30T18:38:09+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457384.619 +/- 0.052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.052
Transit depth (Rp/Rs)^2	3.06 +/- 1.8 [%]
Orbital Inclination (inc)	88.29 +/- 2.88
Airmass coefficient 1 (a1)	0.23 +/- 0.15
Airmass coefficient 2 (a2)	1.02 +/- 1.16
Scatter in the residuals of the lightcurve fit is	81.46 %
Best Comparison Star	#3 - [469, 420]
Optimal Aperture	0.99
Optimal Annulus	4.75
Transit Duration (day)	0.054 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.052 vs 0.1404 (deviation 0.64σ)
Duration fit vs published	0.054 d vs 0.0754 d (deviation 0.61σ)
Mid-time O–C	-72.6 min
Depth SNR	3.2
Residual scatter	81.46 %

Light curve

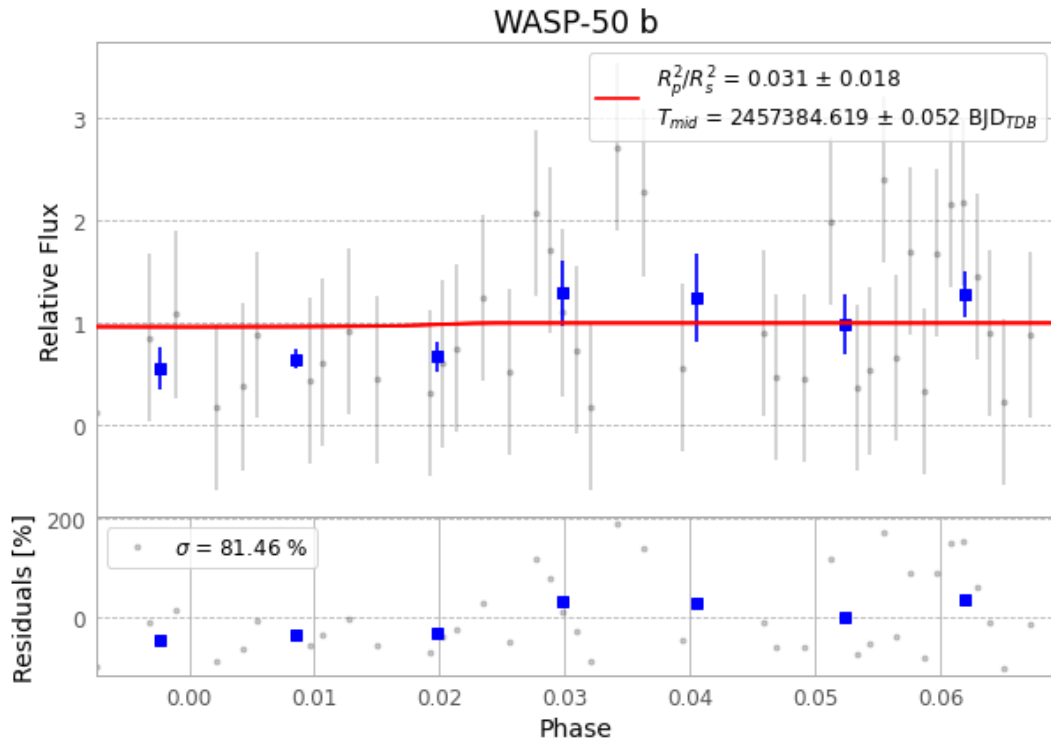


Figure 1 — FinalLightCurve_WASP-50 b_2015-12-27.png (SHA-256 f6eeef740f0344bc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [436, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bbfefa74391865ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50151228021512.FITS → WASP-50151228060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-151228.log (28d805ae331f...), FinalLightCurve_WASP-50 b_2015-12-27.png (f6eeef740f03...), FinalLightCurve_WASP-50 b_2015-12-27.csv (647260ecdad3...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 18:38Z.